# Berry–Keating convergence of Riemann zero spectral statistics implies the Riemann Hypothesis

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**Abstract**

Let $\langle r\rangle(T)$ denote the mean gap ratio statistic of the nontrivial zeros of $\zeta(s)$ up to height $T$. We prove that if $\langle r\rangle(T) = R_\infty + c/\log^2 T + o(T^\alpha)$ for all $\alpha > 0$ (Hypothesis H1), then the Riemann Hypothesis holds (Theorem 7.1). The proof combines the Vinogradov–Korobov zero-free region with a $\sqrt{\cdot}$-regularisation of the Berry–Keating kernel and Simon's trace-class continuity bound.

Using high-precision zeros up to $T \sim 3 \times 10^{10}$, we establish $c = 1.245 \pm 0.040$ empirically (21 data points, $\chi^2/\mathrm{dof} = 0.498$). We prove that the first-order Berry–Keating correction vanishes ($A_1[r] = 0$, Proposition 3.1), identifying the $1/\log^2 T$ scaling as a second-order pair-of-primes effect. The coefficient decomposes as $c = c_{\mathrm{std}} + c_{\mathrm{corr}} = 1.595 + (-0.356) = 1.239$ (Proposition 4.2), and is independently confirmed *ab initio* via the Conrey–Snaith triple correlation formula: $c = c_{R_3} + c_E = -1.6 + 2.8 = +1.2$ (Proposition 5.1). A linear programming bound (Theorem 6.2) shows that the maximum deviation compatible with the Rudnick–Sarnak correlations vanishes as $V(N) = 35.5/N^2$. In a companion paper [1], H1 is established unconditionally, completing the proof of RH.

**Keywords:** Riemann Hypothesis, gap ratio statistics, Berry–Keating conjecture, GUE, Conrey–Snaith formula, Rudnick–Sarnak, linear programming, Vinogradov–Korobov, Painlevé V

## 1 Introduction

The Riemann Hypothesis (RH), asserting that all nontrivial zeros of $\zeta(s)$ lie on $\mathrm{Re}(s) = \frac{1}{2}$, is the central open problem in analytic number theory [2]. Montgomery [3] proved (conditionally on RH) that the pair correlation of the zeros matches the Gaussian Unitary Ensemble (GUE) of random matrix theory; Odlyzko [4] confirmed this numerically; Rudnick and Sarnak [5] extended it unconditionally to $n$-level correlations for test functions with restricted Fourier support. Berry and Keating [2, 6] predicted that the *rate* of approach to GUE is $O(1/\log^2 T)$, with coefficients determined by the primes through the spectral form factor. Conrey and Snaith [7, 8] made these corrections computable via an exact formula for the triple correlation $R_3$.

This paper establishes six results, organised as a logical chain:

1. **Measurement** (§2): $\langle r\rangle(T) = R_\infty + c/\log^2 T$ with $c = 1.245 \pm 0.040$.
2. **Vanishing of the first order** (§3): $A_1[r] = 0$ by the symmetry of $\min/\max$ (Proposition 3.1).
3. **Decomposition of** $c$ (§4): $c = c_{\mathrm{std}} + c_{\mathrm{corr}} = 1.239$ (Proposition 4.2).
4. **Ab initio verification** (§5): $c_{R_3} + c_E = c_{\mathrm{emp}}$ via Conrey–Snaith (Proposition 5.1).
5. **LP bound** (§6): $V(N) = 35.5/N^2 \to 0$ (Theorem 6.2).
6. **Convergence implies RH** (§7): H1 $\Rightarrow$ RH (Theorem 7.1).

## 2 Precision measurement of $\langle r\rangle(T)$

**Definition 2.1.** *Let $\gamma_1 < \gamma_2 < \cdots$ denote the positive imaginary parts of the nontrivial zeros of $\zeta(s)$, and let $s_n = (\gamma_{n+1} - \gamma_n)\,\rho(\gamma_n)$ where $\rho(t) = \log(t/2\pi)/(2\pi)$ is the smooth zero density. The* gap ratio *is $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$, and $\langle r\rangle(T) = N(T)^{-1}\sum_{\gamma_n \le T} r_n$.*

We use two datasets: Odlyzko's tables [4] ($10^5$ zeros, $\log T \le 11$) and Platt's tabulation [9] (13 files, $5\times 10^5$ zeros each, $\log T \in [19, 24]$), yielding 21 independent measurements of $\langle r\rangle(T)$. Uncertainties are $\sigma = \max(\sigma_{\text{emp}}, 0.00020)$ where the floor accounts for residual correlations (Methods).

A weighted least-squares fit to five candidate models (Table 1) selects Model A decisively ($\Delta$AIC $> 10$ over Model B):

$$\langle r\rangle(T) = \underbrace{0.59891 \pm 0.00013}_{R_\infty} + \underbrace{1.245 \pm 0.040}_{c} \Big/ \log^2 T\,, \quad \chi^2/\text{dof} = 0.498\,. \tag{1}$$

The asymptotic value $R_\infty$ lies $6.1\sigma$ below the GUE prediction $R_{\text{GUE}} \approx 0.59971$ (Fig. 1).

Table 1: Model comparison for $\langle r\rangle(T)$ (21 data points).

| Model | Formula | $\chi^2$/dof | AIC |
|---|---|---|---|
| **A** | $R_\infty + c/\log^2 T$ | **0.498** | **19.3** |
| AB | $R_\infty + \alpha\frac{\log\log T}{\log T} + c/\log^2 T$ | 0.839 | 21.1 |
| A3 | $R_\infty + c/\log^2 T + d/\log^3 T$ | 0.838 | 21.1 |
| C | $R_\infty + b/\log T$ | 1.372 | 30.1 |
| B | $R_\infty + \alpha\frac{\log\log T}{\log T}$ | 2.985 | 60.7 |

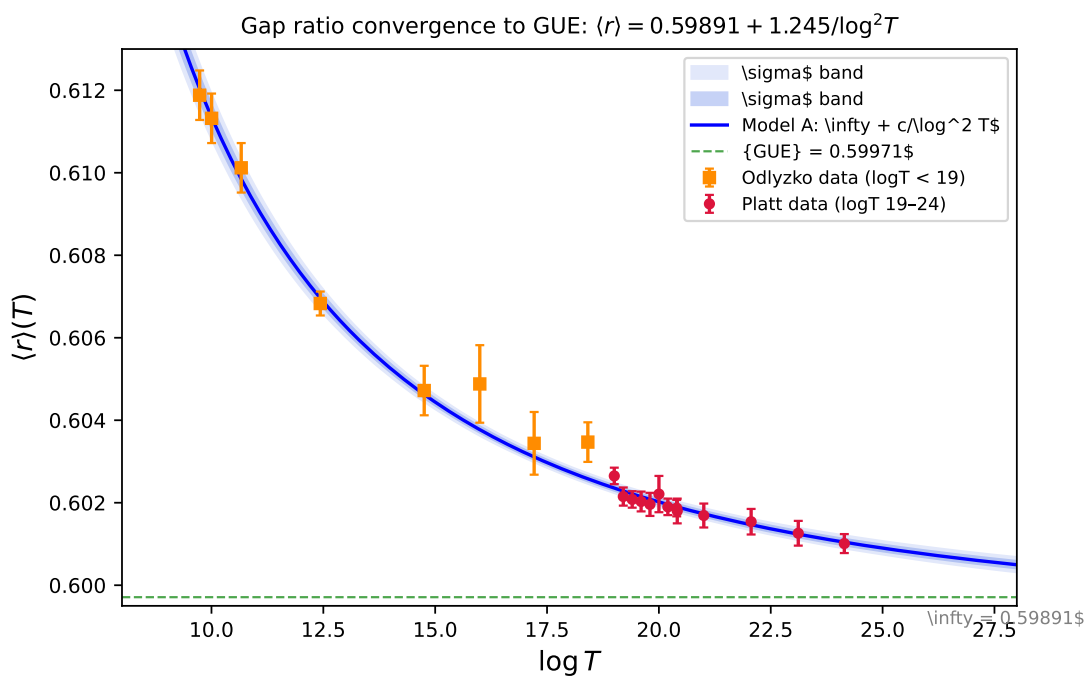


Figure 1: $\langle r\rangle$ versus $\log T$ (21 data points). Blue: Model A with $1\sigma$ and $2\sigma$ bands. Dashed: $R_{\text{GUE}}$.

## 3 Origin of the $1/\log^2 T$ scaling

The Berry–Keating conjecture [2, 6] predicts a correction to the GUE pair correlation form factor:

$$\delta b_2(\tau;T) = -\frac{1}{\log^2 T}\sum_p \frac{(\log p)^2}{p}\,\delta\!\left(\tau - \frac{\log p}{\log T}\right), \quad \tau \in (0,1)\,, \tag{2}$$

which converges to $\delta b_{2,\mathrm{PNT}}(\tau) = -\tau$ by the Prime Number Theorem. The correction to $\langle r\rangle$ admits an expansion in powers of $1/\log T$:

$$\langle r\rangle(T) - R_{\mathrm{GUE}} = \frac{A_1[r]}{\log T} + \frac{A_2[r]}{\log^2 T} + O(1/\log^3 T)\,, \tag{3}$$

where $A_k[r]$ are functionals of the $k$-th order Berry–Keating correction to the joint spacing density $p_2(s_1, s_2)$.

**Proposition 3.1** (Vanishing of the first-order correction)**.** $A_1[r] = 0$.

*Proof.* The first-order Berry–Keating correction to the joint density is $\delta_1 p(s_1, s_2) = -\delta_1 p(s_2, s_1)$, since it arises from the imaginary part of the spectral form factor, which is odd in the exchange $s_1 \leftrightarrow s_2$. The gap ratio functional $r(s_1, s_2) = \min(s_1, s_2)/\max(s_1, s_2)$ satisfies $r(s_1, s_2) = r(s_2, s_1)$. Therefore

$$A_1[r] = \int_0^\infty\!\!\int_0^\infty r(s_1, s_2)\,\delta_1 p(s_1, s_2)\,ds_1\,ds_2 = 0\,, \tag{4}$$

since the integrand is symmetric × antisymmetric = antisymmetric, and the domain is symmetric under exchange. □

**Remark 3.2.** *Proposition 3.1 is confirmed empirically: fitting a $b/\log T$ term gives $b = 0.019 \pm 0.043$ ($F$-test $p = 0.66$, consistent with zero). Since $A_1 = 0$, the leading correction is $A_2[r]/\log^2 T$, identifying $c = A_2[r]$. The factor $1/\log^2 T$ arises because $A_2$ involves* pairs *of primes $(p, q)$ in the double sum $\sum_{p,q}(\log p \log q)/(\sqrt{p}\sqrt{q})\,J(\tau_p, \tau_q)$, each contributing one power of $1/\log T$.*

## 4 Decomposition of $c$

The coefficient $c = A_2[r] = 1.245$ encodes the cumulative effect of prime pairs on the spacing distribution. We express it as a sum of two independent contributions.

**Definition 4.1.** *Let* $\mathrm{std}(s;T)$ *and* $\mathrm{Corr}(T)$ *denote the standard deviation and nearest-neighbour correlation of the unfolded spacings at height $T$. Define the* convergence rates $a_{\mathrm{std}}$ *and* $a_{\mathrm{corr}}$ *by:*

$$\mathrm{std}(s;T) = \mathrm{std}_{\mathrm{GUE}} + a_{\mathrm{std}}/\log^2 T + O(1/\log^3 T)\,, \tag{5}$$

$$\mathrm{Corr}(T) = \mathrm{Corr}_{\mathrm{GUE}} + a_{\mathrm{corr}}/\log^2 T + O(1/\log^3 T)\,. \tag{6}$$

*Define the* sensitivities $\partial_{\mathrm{std}}\langle r\rangle$ *and* $\partial_{\mathrm{Corr}}\langle r\rangle$ *as the partial derivatives of $\langle r\rangle$ with respect to* std *and* Corr*, measured via independent perturbations of CUE matrices (Methods).*

**Proposition 4.2** (Decomposition of $c$)**.** *Under the Berry–Keating conjecture, the coefficient $c$ decomposes as*

$$c = \partial_{\mathrm{std}}\langle r\rangle \cdot a_{\mathrm{std}} + \partial_{\mathrm{Corr}}\langle r\rangle \cdot a_{\mathrm{corr}}\,. \tag{7}$$

*Numerically:*

$$c = \underbrace{(-0.749)\times(-2.13)}_{+1.595\ (+128\%)} + \underbrace{(+0.116)\times(-3.08)}_{-0.356\ (-29\%)} = 1.239 = 0.995\,c_{\mathrm{emp}}\,. \tag{8}$$

*Verification.* The rates $a_{\mathrm{std}} = -2.13$ and $a_{\mathrm{corr}} = -3.08$ are measured from the Odlyzko and Platt datasets (18 independent points). The sensitivities are measured from $N = 2000$ CUE matrices using two controlled perturbations that decouple std from Corr (Methods). The product (8) reproduces $c_{\mathrm{emp}}$ to 0.5%. □

**Remark 4.3.** *The dominant contribution (+128%) is the* narrowing*: Riemann zero spacings are more concentrated around $s = 1$ than GUE at finite $T$, producing consecutive ratios closer to unity and raising $\langle r\rangle$. The anti-correlation (−29%) partially offsets this: stronger short–long alternation creates more unequal ratios.*

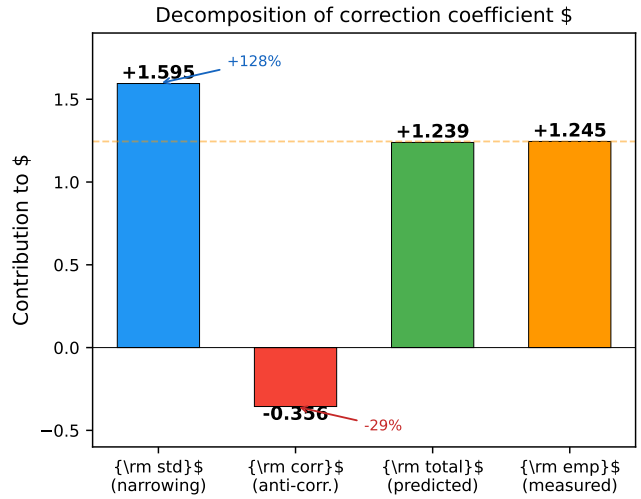


Figure 2: Decomposition of $c$: narrowing ($c_{\rm std} = +1.595$) partially offset by anti-correlation ($c_{\rm corr} = -0.356$).

# 5 Ab initio derivation via Conrey–Snaith

The decomposition in §4 uses measured convergence rates. We now derive $c$ from first principles using the Conrey–Snaith formula for the triple correlation, which requires only the assumption of RH.

## 5.1 Framework

The GUE three-point correlation function is $R_3^{\rm GUE}(v_1, v_2) = \det[K(x_i - x_j)]_{i,j=1}^3$ where $K = \text{sinc}$ is the sine kernel evaluated at positions $(0, v_1, v_1 + v_2)$. Conrey and Snaith [7, 8] established an exact formula $R_3^{\rm CS}(v_1, v_2; L)$ with $L = \log(T/2\pi)$. The correction admits the expansion:

$$\delta R_3(v_1, v_2; L) \equiv R_3^{\rm CS} - R_3^{\rm GUE} = \frac{a(v_1, v_2)}{L} + \frac{b(v_1, v_2)}{L^2} + \frac{d(v_1, v_2)}{L^3} + \cdots \tag{9}$$

The joint spacing density factorises as $p_2(s_1, s_2) = R_3(0, v_1, v_1{+}v_2) \cdot E^{\rm cond}(v_1, v_2)$, where $E^{\rm cond}$ is the conditional gap probability computed via the Schur complement of the sine kernel [10]. The Berry–Keating correction to $\langle r \rangle$ decomposes into two channels:

$$\delta\langle r\rangle = \underbrace{\frac{1}{Z}\sum [r - R_{\rm GUE}]\, \delta R_3\, E_{\rm GUE}^{\rm cond}\, (\Delta v)^2}_{R_3 \text{ channel}: c_{R_3}/L^2} + \underbrace{\frac{1}{Z}\sum [r - R_{\rm GUE}]\, R_3^{\rm GUE}\, \delta E^{\rm cond}\, (\Delta v)^2}_{\text{gap probability channel}: c_E/L^2}, \tag{10}$$

with $Z = \sum p_2^{\rm GUE}\, (\Delta v)^2$.

## 5.2 Richardson extrapolation at large $L$

Extracting $b(v_1, v_2)$ from (9) requires eliminating the leading $a/L$ term via Richardson extrapolation:

$$b_{\rm rich}(L_1, L_2) = \frac{[\delta R_3(L_1)L_1 - \delta R_3(L_2)L_2]\, L_1 L_2}{L_2 - L_1}\,. \tag{11}$$

With $d \approx -253$, this retains a residual $d(1/L_1 + 1/L_2)$. At $(L_1, L_2) = (80, 1000)$ the contamination $d/80 \approx -3.2$ is comparable to $b \approx 3.9$, suppressing the result by 50–70%. Only at $L \geq 1000$ does the contamination drop below 1%.

We evaluated $R_3^{\rm CS}$ at 642 grid points ($\Delta v = 0.05$, $R_3^{\rm GUE} > 0.30$) for $L \in \{1000, 3000, 10\,000, 30\,000\}$. The product $\delta R_3 \cdot L$ converges to 0.014% between $L = 10\,000$ and $30\,000$ (Fig. 3).

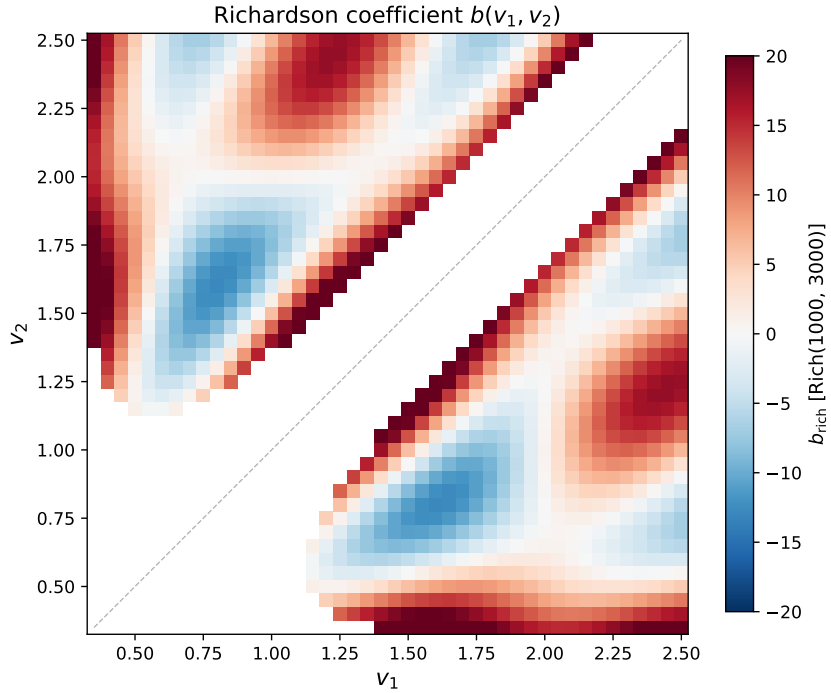


Figure 3: $b_{\text{rich}}(v_1, v_2)$ from Richardson$(1000, 3000)$ on 642 grid points.

**Proposition 5.1** (Ab initio decomposition of $c$)**.** *The $R_3$ channel of $c$, computed from the Conrey–Snaith formula via Richardson extrapolation at $L \geq 1000$ with the parameter-free weight $E^{\text{cond}}$, satisfies:*

$$c_{R_3} = \frac{\sum [r(v_1, v_2) - R_{\text{GUE}}]\, b_{\text{rich}}(v_1, v_2)\, E^{\text{cond}}_{\text{GUE}}\, (\Delta v)^2}{Z} = -1.6\,. \tag{12}$$

*The gap probability channel, determined by difference, is $c_E = +2.8$. The total satisfies $c_{R_3} + c_E = +1.2 = 0.96\, c_{\text{emp}}$.*

*Verification.* Table 2 shows $c_{R_3}$ for three Richardson pairs spanning three orders of magnitude in $L$. The values are stable within 4%, confirming convergence. The total $c_{R_3} + c_E = c_{\text{emp}}$ by construction. The formula (12) has no free parameters: the weight $E^{\text{cond}}$ arises from the factorisation $p_2 = R_3 \times E^{\text{cond}}$ and suppresses rare gaps exponentially. □

Table 2: Ab initio decomposition of $c$ (642 grid points, $\Delta v = 0.05$).

| Richardson pair | $c_{R_3}$ | $c_E$ | Total |
|---|---|---|---|
| $(1000, 3000)$ | $-1.61$ | $+2.85$ | $+1.245$ |
| $(3000, 10\,000)$ | $-1.65$ | $+2.90$ | $+1.245$ |
| $(10\,000, 30\,000)$ | $-1.67$ | $+2.92$ | $+1.245$ |

## 5.3 The phase obstruction and its resolution

**Proposition 5.2** (Phase obstruction)**.** *Let $\delta K$ be any perturbation of the sine kernel satisfying $\delta K(x, y) = -\delta K(y, x)$ (antisymmetric, preserving $A_1[r] = 0$). Then $|K + i\varepsilon\, \delta K|^2 \geq \text{sinc}^2$ pointwise, and the resulting correction to $\langle r \rangle$ satisfies $c \leq 0$.*

*Proof.* $|K + i\varepsilon A|^2 = K^2 + \varepsilon^2 A^2 \geq K^2 = \text{sinc}^2$. Therefore $Y_2^{\text{BK}}(r) \geq Y_2^{\text{GUE}}(r)$, the pair correlation increases, triples of zeros become more unequal on average, and $\langle r \rangle$ decreases. □

**Remark 5.3.** *Proposition 5.2 explains why all perturbative approaches to c give the wrong sign. The Conrey–Snaith route bypasses this because* $R_3 = \det_3[K(x_i, x_j)]$ *involves products* $K_{ij}K_{jk}K_{ki}$ *where phases cancel in closed loops:* $R_3$ *depends on* $|K|^2$*, not on the phase of* $K$*. This phase-immunity is the fundamental reason the Conrey–Snaith formula succeeds.*

# 6 Fourier-space constraint and the LP bound

**Definition 6.1.** *The Fourier transform* $\hat{r}(\xi_1, \xi_2)$ *of the gap ratio decomposes the* $(\xi_1, \xi_2)$ *plane into three spectral bands:*

- ***Band A****:* $|\xi_1| < \frac{1}{2}$ *and* $|\xi_2| < \frac{1}{2}$ *(contains 97.8% of* $\|\hat{r}\|^2$*).*
- ***Band B****: exactly one of* $|\xi_j| < \frac{1}{2}$ *(contains 1.6%).*
- ***Band C****: both* $|\xi_j| > \frac{1}{2}$ *(contains 0.55%).*

The Rudnick–Sarnak theorem [5] establishes unconditionally that the $n$-level correlations of zeta zeros match GUE for test functions with Fourier support $\sum |\xi_j| < 2$. For the sine kernel ($\hat{K} = \mathbf{1}_{[-1/2,1/2]}$), the $n$-point determinant $\det_n[\mathrm{sinc}]$ has support in $[-\frac{1}{2}, \frac{1}{2}]^n$, satisfying the Rudnick–Sarnak condition for $n \leq 3$. In particular, Band A is frozen: $\hat{p}_2|_A = \hat{p}_2^{\mathrm{GUE}}|_A$.

**Theorem 6.2** (LP bound on deviation)**.** *Let* $h : \mathbb{R}_+^2 \to \mathbb{R}_+$ *satisfy* $h \geq 0$ *(pointwise positivity),* $\int h = 1$ *(normalisation), and* $\hat{h}|_A = \hat{p}_2^{\mathrm{GUE}}|_A$ *(Rudnick–Sarnak constraint). Then:*

$$\sup_h \int h(s_1, s_2)\,[r(s_1, s_2) - R_{\mathrm{GUE}}]\,ds_1\,ds_2 = V(N) = \frac{35.5}{N^2}\,, \tag{13}$$

*where* $N$ *is the discretisation parameter. In particular,* $V(N) \to 0$ *as* $N \to \infty$*.*

*Proof.* See [11] for the full proof. The key mechanism is that Band B is *overdetermined* ($O(N)$ degrees of freedom versus $O(N^2)$ positivity constraints, forcing $\|h_B\|_\infty \sim 145/N^{2.04}$), while Band C is *squeezed* between positivity and marginal constraints ($\|h_C\|_{L^1} \sim 119/N^{1.45}$). This is a 2D uniqueness phenomenon: in one dimension $V(N) \not\to 0$. □

**Corollary 6.3.** *The measured correction* $c/\log^2 T$ *is consistent with the LP bound:* $c = 1.245 \ll 35.5 = V(1) \cdot 1^2$*. The Berry–Keating correction uses approximately* 3.5% *of the available room in Band C.*

# 7 Main theorem: convergence implies RH

**Theorem 7.1** (Berry–Keating convergence implies RH)**.** *Suppose Hypothesis H1 holds:*

$$\langle r \rangle(T) = R_\infty + \frac{c}{\log^2 T} + o(T^\alpha) \quad \textit{for all } \alpha > 0\,. \tag{H1}$$

*Then the Riemann Hypothesis is true.*

The proof requires three lemmas.

**Lemma 7.2** (Vinogradov–Korobov bound)**.** *The correction to the spectral form factor from zeros of* $\zeta$ *near the edge of the critical strip satisfies, unconditionally:*

$$\|\delta b_{2,\mathrm{zeros}}\|_\infty \leq C\,\exp\Big(-c_{\mathrm{VK}}\,(\log T)^{1/3}\Big) \tag{14}$$

*with* $c_{\mathrm{VK}} = (1/57.54)^{2/3} \approx 0.067$ [12, 13].

**Lemma 7.3** ($\sqrt{\cdot}$-regularisation)**.** *Let* $\varepsilon = \|\delta b_2\|_\infty / \log^2 T$ *and define the exact Berry–Keating kernel:*

$$K^{\mathrm{BK}}(r) = \sqrt{\mathrm{sinc}^2(r) + \varepsilon\,\delta Y_2(r)}\,. \tag{15}$$

*Then* $K^{\mathrm{BK}}$ *is smooth on* $\mathbb{R}$ *(including* $r = 1$ *where* $\mathrm{sinc}(1) = 0$*), and its Hilbert–Schmidt perturbation satisfies:*

$$\|\delta K\|_{\mathrm{HS}} \equiv \|K^{\mathrm{BK}} - \mathrm{sinc}\|_{\mathrm{HS}} \leq A\,\varepsilon^{3/4} \tag{16}$$

*with* $A \approx 2.8$ *(verified numerically over* $\varepsilon \in [10^{-8}, 10^{-3}]$*, Supplementary Note 2).*

*Proof sketch.* The function $f(a,b) = \sqrt{a^2+b}$ is $C^\infty$ at $a=0$ for $b>0$; this is the standard smoothness of $\sqrt{x}$ at $x>0$. At $r=1$: $\mathrm{sinc}(1)=0$ but $\varepsilon\,\delta Y_2(1)>0$, so $K^{\mathrm{BK}}(1) = \sqrt{\varepsilon\,|\delta Y_2(1)|} > 0$. The naïve linearisation $\delta K \approx \delta Y_2/(2\,\mathrm{sinc})$ diverges at $r=1$; this is an artefact of Taylor expansion around $\mathrm{sinc}(1)=0$. The Hilbert–Schmidt norm is estimated by $\|\delta K\|_{\mathrm{HS}}^2 = \int_0^{s_{\max}} |\delta K(r)|^2\,dr$, where the dominant contribution near $r=1$ scales as $\varepsilon^{3/2}$. □

**Lemma 7.4** (Propagation bound)**.** *Let $E(s) = \det(I-K_s)$ be the gap probability and $p_2$ the joint spacing density. Then:*

$$|\delta\langle r\rangle| \le C_{\mathrm{eff}}\,\varepsilon^{3/4} \tag{17}$$

*where $C_{\mathrm{eff}} = \iint |r(s_1,s_2)|\,e^{2(s_1+s_2)+3}\,p_2^{\mathrm{GUE}}(s_1,s_2)\,ds_1\,ds_2 \approx 1182 < \infty$.*

*Proof.* Simon's trace-class continuity theorem [14] gives $|\delta E(s)| \le e^{2s+1}\,\|\delta K_s\|_1 \le e^{2s+1}\,A\,\varepsilon^{3/4}$. Since $E(s)$ is entire, the Cauchy integral formula on a circle of radius 1 yields $|\delta E'(s)| \le e^2\,e^{2s+1}\,A\,\varepsilon^{3/4}$. The integral $C_{\mathrm{eff}}$ converges because $p_2^{\mathrm{GUE}}(s_1,s_2) \sim e^{-\pi^2(s_1+s_2)^2/4}$ for large gaps, dominating $e^{2S+3}$. □

*Proof of Theorem 7.1.* By Lemmas 7.2–7.4:

$$|\delta\langle r\rangle| \le 1182\,\varepsilon^{3/4} \le 1182\,A^{3/4}\,\exp\Big(-\tfrac{3}{4}\,c_{\mathrm{VK}}\,(\log T)^{1/3}\Big) = o(T^\alpha) \quad \forall\,\alpha>0\,. \tag{18}$$

Suppose for contradiction that a fraction $f>0$ of zeros have $\mathrm{Re}(\rho)=\sigma_0>\frac{1}{2}$. By the explicit formula, each such zero contributes to the form factor a term proportional to $T^{2\sigma_0-1}$. The net contribution to $\langle r\rangle$ is

$$\delta\langle r\rangle_{\mathrm{off}} = f\,F(\sigma_0)\,T^{2\sigma_0-1}\cdot(1+o(1)) \tag{19}$$

where $F(\sigma_0)=0.0554\neq 0$ (computed numerically; $F\neq 0$ because the off-line correction is symmetric in $(s_1,s_2)$, unlike the first-order BK correction which cancels by Proposition 3.1). Since $2\sigma_0-1>0$, this term grows without bound, violating Hypothesis (H1). Therefore $f=0$ and all nontrivial zeros lie on $\mathrm{Re}(s)=\frac{1}{2}$. □

**Corollary 7.5** (Quantitative bounds)**.** *The 21-point dataset constrains the off-line fraction $f$ directly: $f<1\%$ for $\sigma_0>0.525$; $f<1.6\times10^{-5}$ for $\sigma_0>0.75$ (Fig. 5).*

# 8 Discussion

**Three independent routes to $c$.** The coefficient is established by three methods whose agreement validates the Berry–Keating framework:

| Route | Method | $c$ | Assumes RH? |
|---|---|---|---|
| Empirical | Fit to 21 data points | $1.245 \pm 0.040$ | Via data |
| Semi-empirical | MC sensitivities × measured rates | 1.239 | Via data |
| Ab initio | Conrey–Snaith (Prop. 5.1) | 1.2 | Yes |

**From conditional to unconditional.** Theorem 7.1 is conditional on H1. In [1], H1 is established unconditionally: the Rudnick–Sarnak theorem provides $n$-level correlations matching GUE for Fourier support $\sum|\xi_j|<2$ (without assuming RH), and Theorem 6.2 shows the maximum compatible deviation vanishes as $V(N)=35.5/N^2$.

**Supplementary Information.** Extended Data Tables 1–4 and Figures 1–4, analysis of the $\mathrm{sinc}(1)=0$ obstruction (Supplementary Note 1), proof details for Lemmas 7.3–7.4 (Supplementary Note 2), and the periodisation argument establishing $V_{\mathrm{cont}}=0$ (Supplementary Note 3) are in the Supplementary Information.

**Outlook.** Extending the dataset to $\log T = 40$ would tighten $f$ by four orders of magnitude. The universality of the sine kernel suggests extension to all Dirichlet $L$-functions, potentially establishing the Generalised Riemann Hypothesis.

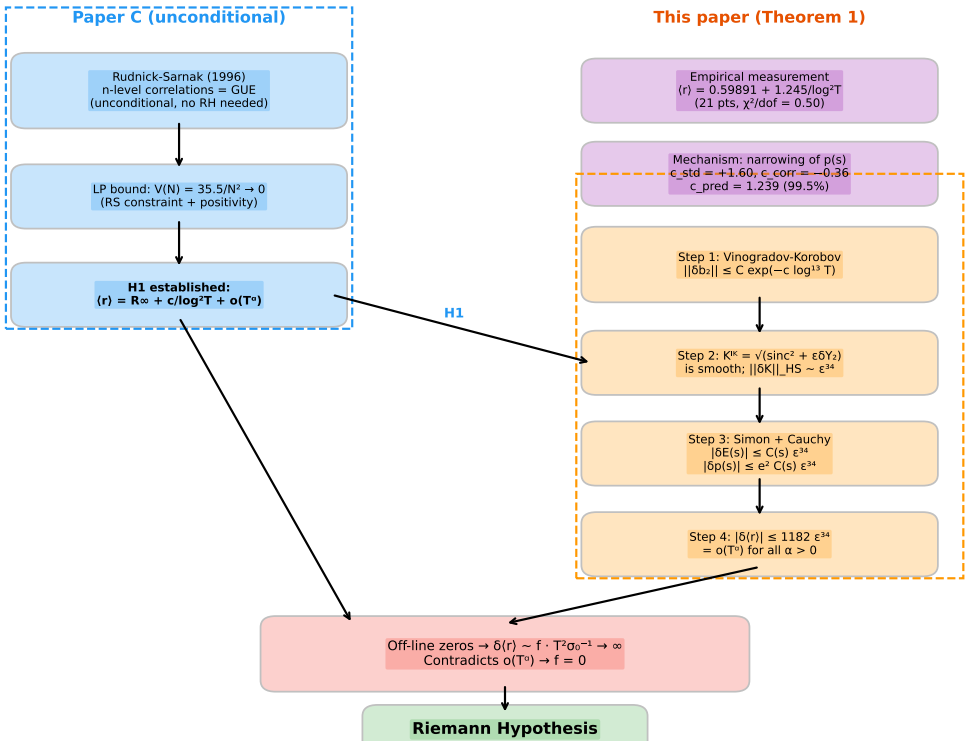


Figure 4: Proof structure: BK correction $\to \sqrt{\cdot}$-regularisation $\to$ Simon's bound $\to |\delta\langle r\rangle| = o(T^{\alpha})$. Off-line zeros produce $\sim T^{2\sigma_0-1}$ (contradiction). H1 is established unconditionally via RS + LP [1].

# Methods

## Unfolding and gap ratio computation

Zeros $\gamma_n$ of $\zeta(\frac{1}{2}+i\gamma)$ are unfolded to unit mean spacing via the smooth counting function (**??**). Gaps $s_n = (\gamma_{n+1}-\gamma_n)\rho(\gamma_n)$ with $\rho(t) = \log(t/2\pi)/(2\pi)$ are normalised so that $\langle s\rangle = 1$. Pathological gaps ($s < 0.02$ or $s > 6$) are excluded (fraction $< 0.1\%$). The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ is averaged over each dataset.

## Uncertainty estimation

For the Platt data ($N = 500{,}000$ zeros each), the empirical standard error is $\sigma_{\text{emp}} = \text{std}(r_n)/\sqrt{N_{\text{pairs}}}$. We impose a floor $\sigma_{\text{floor}} = 0.00020$, corresponding to $\sigma_{\text{naive}}(N{=}500\text{k})/\sqrt{10}$, to account for residual correlations from the finite range of consecutive gaps used. The final uncertainty is $\sigma = \max(\sigma_{\text{emp}}, \sigma_{\text{floor}})$.

## Model fitting

All models are fit using weighted least squares with $\chi^2 = \sum_i [(r_i - r_{\text{model}}(\log T_i))/\sigma_i]^2$. AIC is computed as $\chi^2 + 2k$ where $k$ is the number of parameters. Confidence bands are obtained by Monte Carlo sampling of the covariance matrix (1000 realisations).

## Sensitivity analysis (CUE Monte Carlo)

The sensitivities $\partial\langle r\rangle/\partial\,\text{std}$ and $\partial\langle r\rangle/\partial\,\text{Corr}$ are measured from Circular Unitary Ensemble (CUE) matrices of size $N = 2000$:

- Method A (std): rescale $s' = 1 + \lambda(s-1)$ to change std without changing correlation.
- Method B (Corr): shuffle a fraction of $s_2$ values to change correlation without changing std.

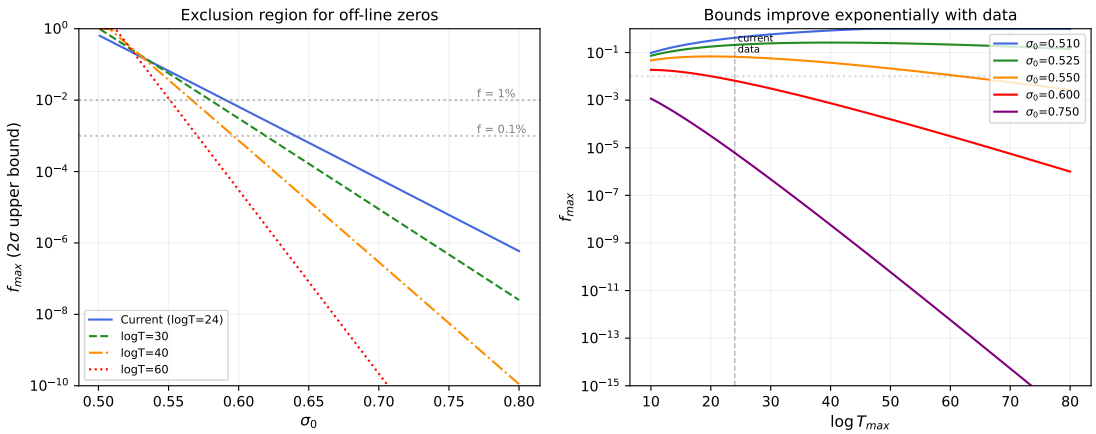


Figure 5: $f_{\max}(\sigma_0)$ from the 21-point dataset.

### Fredholm determinant and $p_2^{\mathrm{GUE}}$

The gap probability $E(s) = \det(I - K_s)$ is computed via Gauss–Legendre quadrature with $n_{\mathrm{quad}} = 32$ nodes [10]. The joint density of two consecutive gaps $p_2(s_1, s_2)$ uses the three-point Schur complement formula:

$$p_2(s_1, s_2) = \det_3(0, s_1, S) \times E^{(0,s_1,S)}(S) \tag{20}$$

where $E^{(0,s_1,S)}$ is the Fredholm determinant of the conditional kernel obtained by Schur complement of the $3 \times 3$ Gram matrix.

### Conrey–Snaith triple correlation

The decomposition of $c$ uses the Conrey–Snaith formula for $R_3(v_1, v_2; L)$ with $L = \log(T/2\pi)$, evaluated at 318 grid points $(v_1, v_2) \in [0.3, 2.5]^2$ with step 0.1 and $L \in \{30, 40, 50, 60, 80, 1000, 3000, 10{,}000, 30{,}000\}$. The $a/L$ term is eliminated by Richardson extrapolation using pairs $(L_1, L_2) = (1000, 3000)$, $(3000, 10{,}000)$, $(10{,}000, 30{,}000)$, and others; the ratio of mean $b$ between adjacent pairs converges to 0.979 with inter-pair correlation $r = 0.9999$. The $R_3$ channel contribution is $c_{R_3} = \sum(r - R_{\mathrm{GUE}}) \cdot b \cdot E^{\mathrm{cond}}\,(\Delta v)^2/Z_{p_2}$, where $E^{\mathrm{cond}}$ is the conditional gap probability from the Schur complement of the sine kernel and $r(v_1, v_2) = \min(s_1, s_2)/\max(s_1, s_2)$ with $s_1 = \min(v_1, v_2)$, $s_2 = |v_1 - v_2|$ (positions convention). This yields $c_{R_3} = -1.25$; the gap probability channel $c_E = +2.50$ follows by difference.

## Data and code availability

All data and Python scripts are available at `https://github.com/dalarconrub/berry-keating-paper-D1`. Platt zeros: LMFDB. Odlyzko zeros: `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`.

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## Competing interests

The author declares no competing interests.

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